ODP – Client Proposal

Team: *No Fly Zone* **Client(s):** Cornell CALS Extension / E&J Gallo Winery / National Grape

Problem statement

The spotted lantern fly (SLF) is an invasive species in North America. Economic losses to the grape industry in Finger Lakes alone is estimated to reach $1.5 million in the first year of infestation, and double annually. Infested grapevines use 38% less water due to lower sap flow rates, leading to lower quality product. The SLFs also remain on grapevines during the harvesting process, contaminating the grapes and further decreasing yields. Currently, SLF removal is only done at harvest and product processing.

Impact

Passive SLF population control during the growing season will lower total vine damage, and limit the contamination levels for future SLF removal efforts during harvesting.

Proposed direction

Concept: "The Lure"

What it is: A device hung from a trellis; SLFs are first attracted, and then contained.

Minimum viable product / proof-of-concept: A plastic bottle filled with wintergreen oil. A rubber valve would replace the cap and act as a one-way entrance.

Potential long term product: A dual-attraction system of wintergreen oil and a 60hz speaker. A helical wall will rotate and guide SLFs down into a replaceable containment chamber.

Key risks / unknowns

- Cost — Targeting every grapevine would require many lures, what does this add too and and how much is too much?
- Efficacy — Testing how well lures attract is difficult in the classroom.
- Maintenance — How to balance required manual labor and lure effectiveness.

Questions for the client

1. What financial investment would be available for the full-scale use of this product?
2. How much labor would be available for maintaining this product?
3. What level of attachment between a lure and the trellis is required to maintain integrity throughout the growing and harvesting seasons?

References

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